
Optimized Modeling and Predictive Analysis of Complex Systems

Summary

Problem Restatement. In one or two sentences, state the **core question** and **constraints** in your own words; identify **decision variables**, **objectives**, and **feasible region**.

Uniqueness of the Modeling Approach. Briefly state the **model class** (e.g., differential equations, optimization, stochastic simulation) and **distinct assumptions** or **innovations** relative to existing methods.

Core Algorithm Logic. Outline the **solution pipeline**: inputs, main iterations or recurrences, and output metrics; name **software/algorithms** when relevant (e.g., Gurobi, Monte Carlo).

Most Impactful Quantitative Conclusions. Give concrete **numerical results** and **policy implications** (e.g., “cost reduced by 15.4%” or “service level above 98%”); avoid vague phrases. Optionally state **sensitivity** or **robustness** findings.

The four blocks above use bold terms and short paragraphs for modular scannability, with paragraph spacing for clarity.

Keywords: Mathematical Modeling; Sensitivity Analysis; AI Tools Disclosure.

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1 Introduction

1.1 Background and Motivation

Describe the real-world context provided by the problem statement.

AI use disclosure. Our use of AI tools is disclosed in this paragraph (in-text) and in full in the **Report on Use of AI Tools** at the end of this paper.

1.2 Literature Review

Cite existing research to build model credibility.[?]

1.3 Restatement of the Problem

Define specific goals using your own academic language.

2 Assumptions and Notations

2.1 General Assumptions

List simplifying assumptions and their scientific justifications.

2.2 Notation Definition

Use a structured table to define variables for cognitive efficiency.

Table 1: Primary Notations Underpin All Model Equations in This Paper

Variable	Description	Dimension/Units
$P(t)$	Population density at time t	ind/km^2
η	Efficiency of the transmission channel	Dimensionless

3 Data Sources

3.1 Data Description and Task DNA Source

We extract task statements and task ratings for SOC 15-1252.00 from the O*NET 30.1 text database, and we compute task weights from Importance (IM) and Frequency of Task (FT) ratings.[?]

Table 2: Data Sources and Variables for Task DNA

Variable/Data	Description	Source
O*NET Task Statements	Task text and identifiers for SOC 15-1252.00	O*NET 30.1
O*NET Task Ratings (IM, FT)	Importance and Frequency ratings used to compute task weights	O*NET 30.1
Task DNA CSV	Processed task table with IM, FR, weight, and rule-based s/c	Derived from

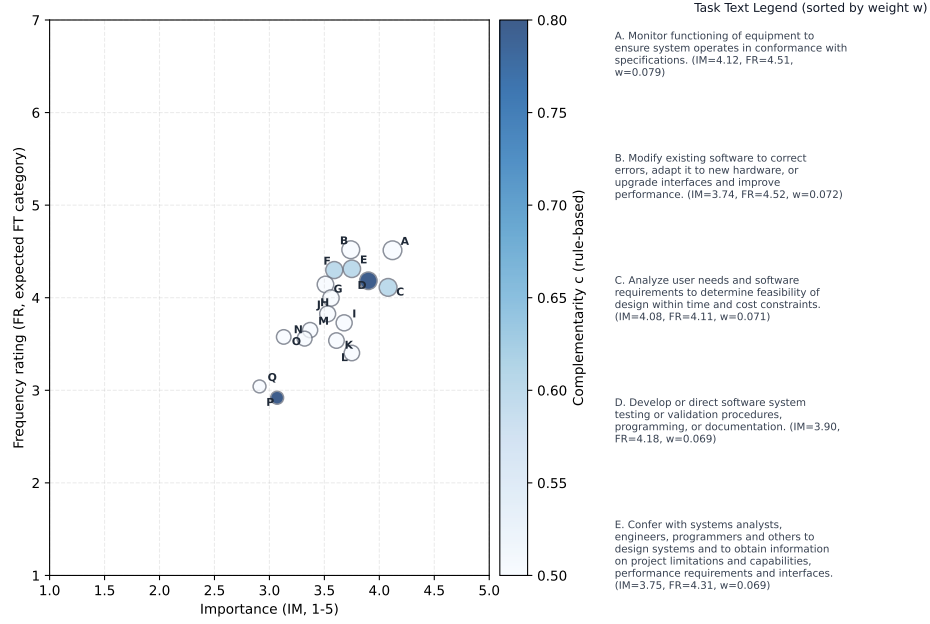


Figure 1: Task-level IM–FR exposure for Software Developers (SOC 15-1252.00). Bubble size encodes normalized task weight w , color encodes rule-based complementarity c , and edge thickness reflects rule-based substitution s . Letter tags map to the task text legend panel.

4 Model Construction and Implementation

4.1 Career Selection as a Modeled Decision

A prerequisite of the problem requires selecting one career each from the STEM, Trade, and Arts sectors. Rather than treating this step as an ad hoc assumption, we explicitly model career selection as an optimization problem. This ensures that subsequent analyses of AI-driven impacts and educational recommendations are grounded in a reproducible and data-informed framework.

4.1.1 Modeling Rationale

The core objective of career selection is not representativeness of all occupations, but *explanatory contrast*. Specifically, the selected careers should differ substantially in how generative artificial intelligence (GenAI) interacts with their task structures, while remaining empirically tractable and policy-relevant.

Based on the problem statement and data availability constraints, we identify three dominant dimensions that govern heterogeneity in AI impact across occupations:

- **Digitizability:** the extent to which core tasks can be represented, automated, or augmented digitally;
- **Physical anchoring:** the degree to which task execution requires on-site physical presence or manual interaction with the environment;

- **Intellectual property and attribution risk:** the sensitivity of value creation to authorship, originality, and ownership.

Each candidate career is represented by a normalized feature vector

$$f(x) = (\text{digit}(x), \text{physical}(x), \text{iprisk}(x)) \in [0, 1]^3, \quad (1)$$

where higher values indicate stronger presence of the corresponding characteristic.

In addition to mechanistic contrast, the problem explicitly emphasizes data-informed reasoning. Therefore, we introduce a data availability score $D(x)$ that reflects whether reliable data sources exist for employment trends, task descriptions, compensation, and educational pathways associated with career x . These data sources include occupational databases [?], post-secondary employment outcomes [?], educational institutional data [?], and occupational classification crosswalks [?].

4.1.2 Objectification of Career Feature Parameters Based on O*NET

To avoid subjective assignment of the career feature parameters $\text{digit}(x)$, $\text{physical}(x)$, and $\text{iprisk}(x)$, we adopt the **O*NET** occupational database maintained by the U.S. Department of Labor as a unified data source, with detailed data provenance summarized in Section 3. O*NET provides a large number of *numerical scale descriptors* based on expert surveys and practitioner feedback for each occupation, with most indicators naturally given on a continuous scale of 0–100, suitable for direct use in quantitative modeling.

Parameter Construction Logic We do not attempt to find the “unique correct” indicator for each dimension, but rather follow the *shortest defensible path* principle: for each feature axis, we select a small number of semantically directly corresponding and downloadable O*NET descriptors and use them as proxy variables for that axis.

- **Digitizability $\text{digit}(x)$** This dimension characterizes the extent to which occupational tasks rely on computers and electronic information flows. We select work activity and work context descriptors from O*NET that are directly related to “working with computers” and “frequency of electronic communication”. The higher the value, the more the occupation is centered on digital media.
- **Physical anchoring $\text{physical}(x)$** This dimension characterizes the degree to which an occupation depends on on-site operations, manual control, and physical posture constraints. We use work context descriptors that reflect “time spent performing hands-on activities” and “physical posture burdens such as kneeling, squatting, or bending”, to quantify whether the occupation requires physical presence and physical operations.
- **Intellectual property / attribution risk $\text{iprisk}(x)$ (proxy indicator)** This dimension does not directly measure the probability of legal litigation, but rather serves as a *proxy indicator* to characterize the dependence of occupational output on original expression and creative contribution. We select ability and work activity descriptors related to “Originality” and “Thinking Creatively” from O*NET. The higher the intensity of original expression, the more likely it is to trigger copyright, attribution, or style attribution disputes when generative models intervene.

Normalization and Reproducibility For each occupation x , let $v_m(x)$ denote its raw rating (0–100) under O*NET descriptor m . We apply min-max normalization across the entire occupation set:

$$z_m(x) = \frac{v_m(x) - \min_y v_m(y)}{\max_y v_m(y) - \min_y v_m(y)} \in [0, 1], \quad (2)$$

and take the equally weighted average of multiple descriptors under the same feature axis to obtain $digit(x)$, $physical(x)$, and $iprisk(x)$. This process is entirely based on publicly available data and explicit rules, ensuring the reproducibility and auditability of parameter values.

We visualize the three-dimensional heterogeneity and its 2D projections to validate that the selected careers occupy distinct regions in the feature space (Fig. 2–3).

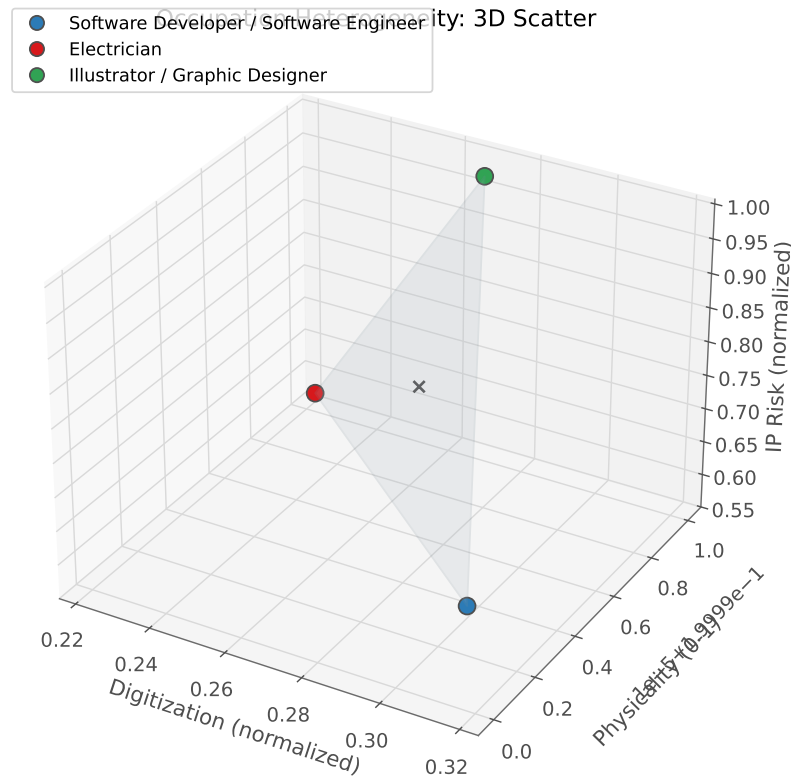


Figure 2: 3D heterogeneity of the selected STEM, Trade, and Arts occupations based on O*NET descriptors (global normalization).

4.1.3 Optimization Formulation

For a triplet of careers (x_1, x_2, x_3) drawn respectively from STEM, Trade, and Arts categories, we define the pairwise contrast between two careers as the Euclidean distance between their feature vectors:

$$\Delta(x_i, x_j) = \|f(x_i) - f(x_j)\|_2. \quad (3)$$

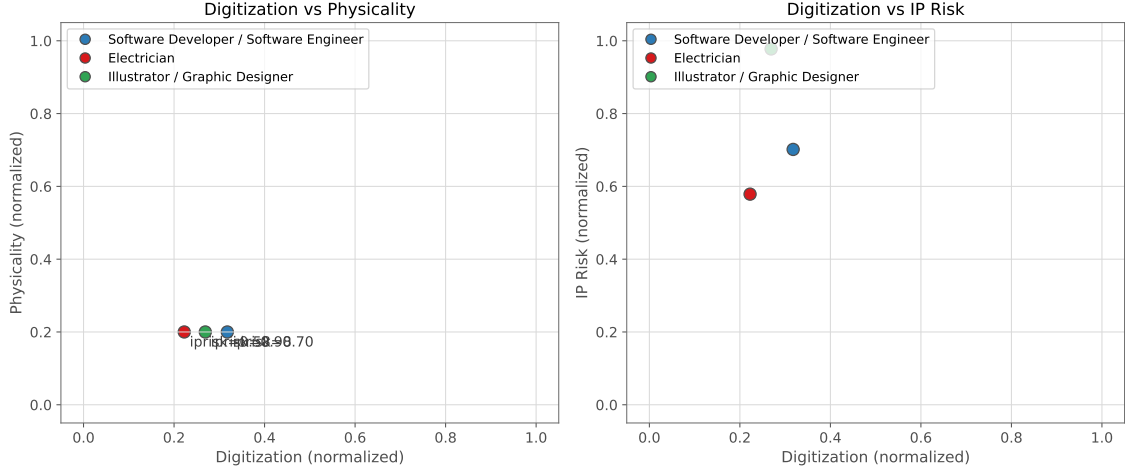


Figure 3: 2D projections of the heterogeneity space; the annotations report the remaining axis value for interpretability (global normalization).

The overall selection objective is then formulated as

$$\max_{x_1 \in STEM, x_2 \in Trade, x_3 \in Arts} \left(\sum_{i < j} \Delta(x_i, x_j) + \lambda \sum_{k=1}^3 D(x_k) \right), \quad (4)$$

where λ balances explanatory contrast against empirical feasibility.

This formulation does not claim global optimality across all possible occupations. Instead, it yields a deliberately constructed high-contrast career set that maximizes analytical clarity under realistic data and time constraints.

4.1.4 Revision and Final Configuration of Selected Careers

In the initial scheme, we selected Structural/Civil Engineer as the STEM occupation. However, upon further examination of the distribution of the three occupations across the “task structure” and “physical constraint” dimensions, we found that this choice exhibits some similarity with the Trade occupation (Electrician) along the “physical anchoring strength” axis, potentially weakening the contrast among the three occupations in the mechanism space.

To maximize task structure heterogeneity and enhance the identifiability of subsequent GenAI impact mechanisms, we revise the STEM endpoint by replacing it with **Software Developer / Software Engineer**.

This revision does not alter the structure of the original selection model, but merely adjusts the position of the occupational endpoint in the feature space: software developers possess extremely high digitizability and extremely low physical anchoring, thereby significantly increasing their distance from electricians in the feature vector $f(x)$ and improving the value of the overall contrast objective function.

It must be emphasized that we do not assume that the software development occupation will be simply “replaced” by GenAI. On the contrary, this occupation is more likely to undergo **task struc-**

ture reorganization: the proportion of coding and boilerplate implementation decreases, while the proportion of tasks such as requirement clarification, system architecture design, verification testing, and security and compliance responsibilities increases. It is precisely this shift from “execution-oriented tasks” to “verification and responsibility-oriented tasks” that makes software development an ideal STEM endpoint for studying GenAI impact mechanisms.

After completing this revision, the final configuration of the three selected occupations is as follows:

- **STEM: Software Developer / Software Engineer.** High digitizability, low physical anchoring; GenAI can directly intervene in core tasks, but primarily manifests as task redistribution rather than occupational extinction.
- **Trade: Electrician.** High physical anchoring, strong on-site constraints; AI primarily serves as a planning, diagnostic, and auxiliary tool, difficult to replace core operational tasks.
- **Arts: Commercial Illustrator / Graphic Designer.** High intellectual property and attribution sensitivity; GenAI directly affects output forms and value pricing mechanisms.

4.2 Model Selection for Occupational Evolution

The second core requirement of the problem is to construct a *data-informed* model of future occupational evolution. Before introducing formal equations, we first define what “future evolution” means in computable terms and then systematically evaluate alternative modeling approaches, so that model choice itself is not an implicit assumption.

4.2.1 Computable Definition of Future Evolution

In this paper, the future evolution of occupation i is characterized by two primary outputs:

- **Occupational demand:** the time evolution of job demand $E_i(t)$ over a 10-year horizon, under fast, medium, and slow GenAI adoption scenarios;
- **Task structure:** the reallocation of task weights $T_i(t)$ within the occupation, indicating which tasks expand, contract, or are reorganized.

This definition ensures that occupational-level outcomes are derived from task-level mechanisms rather than imposed directly.

4.2.2 Comparison of Candidate Modeling Approaches

We consider three representative approaches for modeling occupational evolution:

Approach A: Task-aggregate impact model This approach treats an occupation as a weighted bundle of tasks. GenAI-driven substitution and augmentation effects are specified at the task level and then aggregated to the occupational level using task weights. Most parameters can be constructed from O*NET task data [?] and publicly available AI exposure rubrics [?], providing transparency and reproducibility.

Approach B: Job-posting time-series models This approach infers occupational change from trends in AI-related keywords and skill requirements in job postings. While closely tied to market signals, it suffers from identification challenges and high data-processing overhead, making it unsuitable as the main modeling backbone.

Approach C: Agent-based simulations Agent-based models simulate strategic interactions among firms, workers, and educational institutions. Despite their narrative richness, they require extensive calibration and introduce substantial model complexity, which conflicts with transparency requirements under strict page limits.

4.2.3 Final Model Choice

Based on the above comparison, we adopt the task-aggregate impact model (Approach A) as our primary framework. This choice is motivated by:

- **Mechanistic clarity:** occupational change emerges explicitly from task-level effects;
- **Data alignment:** key parameters are grounded in publicly available occupational data [? ?];
- **Controlled complexity:** the model remains interpretable and suitable for scenario and sensitivity analysis;
- **Transferability:** the framework can be readily extended to other occupations or educational pathways [?].

Approach B is reserved for potential validation or supplementary analysis, while Approach C is excluded from the main analysis due to calibration and transparency concerns.

4.3 Task-Aggregate Dynamic Model of Occupational Evolution

Having justified the task-aggregate approach, we now present its mathematical formulation. We adopt a minimal task-to-occupation framework: GenAI adoption $A(t)$ enters as an exogenous driver, substitution and augmentation are specified at the task level, and occupation-level net impact is derived by task-weight aggregation, yielding a dynamic trajectory for occupational demand $E_i(t)$.

4.3.1 Task Aggregation and Occupational Dynamics

Let occupation i consist of tasks $j = 1, \dots, J_i$ with weights w_{ij} such that

$$\sum_{j=1}^{J_i} w_{ij} = 1. \quad (5)$$

For each task j , let $s_{ij} \in [0, 1]$ denote substitution intensity and $c_{ij} \in [0, 1]$ denote augmentation (complementarity) intensity. Define occupation-level aggregate effects as

$$\alpha_i = \sum_{j=1}^{J_i} w_{ij} s_{ij}, \quad \beta_i = \sum_{j=1}^{J_i} w_{ij} c_{ij}. \quad (6)$$

Let $E_i(t)$ be employment demand for occupation i at time t , and let g_i capture baseline growth/decay from non-AI factors (either estimated from public trends or set to 0 under scenario analysis). Given a scenario family of adoption trajectories $A(t) \in [0, 1]$ (fast/medium/slow), we model occupational evolution as

$$\frac{dE_i(t)}{dt} = E_i(t) \left(g_i + A(t) (\beta_i - \alpha_i) \right). \quad (7)$$

This formulation ensures that occupational outcomes are derived from task-level mechanisms, supporting interpretability, reproducibility, and data-informed reasoning.

5 Simulation Results and Sensitivity Analysis

Provide visual data through plots. Discuss model robustness. Sensitivity analysis is crucial for winning Meritorious or Outstanding awards.

6 Discussion: Strengths and Weaknesses

6.1 Strengths

High predictive accuracy; adaptable to high-dimensional data.[3, 24]

6.2 Weaknesses

Computational complexity for large-scale matrices.

7 Conclusion

Summarize findings and provide actionable insights.

Appendices

Appendix A Python Implementation Code

```
1 def model_solver(data):
2     # Core mathematical logic here
3     return results
```

Report on Use of AI Tools

1. Dialogue Traceability

For each use, provide the exact **Prompt/Query** and the **raw AI Output** (or a representative excerpt). Omit full logs only for translation-only use; in that case, state the **tool name and version**.

- **Example—Prompt:** “Suggest an outline for the model section.” **Output (excerpt):** “1. State assumptions; 2. Define notation; 3. Derive main equations...”
- **Translation-only teams:** Declare tool name and version (e.g., DeepL 4.2, ChatGPT-4); no full prompt log required.

2. Phased Declaration

State in which phases AI was used, e.g.:

- **Initial conception / brainstorming**
- **Code correction / debugging**
- **Translation / copy-editing**

Mark each phase as used or not, and name the tool per phase.

3. Risk Verification

Describe how the team checked for **hallucinations** and **wrong citations**: e.g., manual fact-check of key claims, verification of references, cross-check of formulas and code. If only used for translation, state that no substantive technical content was delegated to AI.